GrideX

Energy. Data. Control.

Complete product and functional specification

ONE PLATFORM	UNIFIED ENERGY MODEL
Sites and assets	Batteries, inverters, charging stations and loads
Infrastructure	Edge controllers, metering and secure communication
Data and services	Prices, charts, analytics and forecast models
Modes and economics	Strategies, constraints, costs and energy balance

Edition 4.0 | English

Full functional scope, including project options. Activated modules, integrations and parameters are defined by the customer configuration and contract.

For energy-site owners, organisation administrators, operators and integrators.

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01 / FULL PRODUCT SCOPE

Platform and product scope

Customer guide

GrideX brings energy assets, local infrastructure, market data and operating rules into a common monitoring and management environment. The full product scope includes core functions, configurable services and project-specific extensions.

Element / setting	Description and use
Sites	Locations, assets, infrastructure and access.
Energy assets	Batteries, inverters, chargers and loads.
Services	Prices, charts, analytics, weather and forecasts.
Mode	Strategies, schedules, limits and alarms.
Administration	Organisations, users, roles, services and settings.

Practical use

This is a functional specification of the full product scope, not a list of services activated in a particular account. Project options form part of the agreed configuration. Equipment, integrations, permissions and the contract determine applicable functions. Example values are illustrative.

02 / FULL PRODUCT SCOPE

Getting started: invitation, sign-in and profile

Settings / Profile

Access starts with an email invitation. The recipient verifies the address and sets a password on the protected identity screen. After sign-in, the portal loads their organisation, roles and authorised Sites.

Element / setting	Description and use
Identity	Email, first and last name; explicit organisation membership.
Sign-in	Email routes to the correct organisation and passes to the password screen.
Recovery	Forgot-password process through the registered email.
Preferences	Language, time zone, units, theme and notifications.
Sign-out	Clear personal context and return to the public demo.

Practical use

An accepted registration invitation does not require a second acceptance button after authentication. On a shared computer, changing person means signing out and signing in again. The demo is separate from customer data and illustrates the product scope.

03 / FULL PRODUCT SCOPE

Navigation and workspaces

All sections

Navigation follows identity, role, services and available assets. Administrative actions are separate from operational measurements, while each Site retains a common context across screens.

Element / setting	Description and use
Overview	Power, energy flows, status and key indicators.
Sites	List and detail with assets, services and infrastructure.
Energy assets	Battery; Inverter; Charging station; Consumer and load.
Infrastructure	One section for controllers, meters, sensors, routers and connectors.
Services / Mode	Personal services; logic, schedule and alarm.
Settings	Users; Plan and subscription; Market; Profile and Documentation.

Practical use

In a real account, asset-type subsections follow accessible inventory. Temporarily offline equipment remains in the list. The demo exposes all categories. Contextual help links to the relevant customer-documentation topic.

04 / FULL PRODUCT SCOPE

Organisations, users and permissions

Settings / Users

Each organisation has an isolated user and data context. Access is specified separately for person, Site, action and service. A role defines permissions; Site assignment determines where they apply.

Element / setting	Description and use
Platform administrator	Manages organisations, catalogue and organisation grants; no self-approval for personal services.
Organisation administrator	Invites people, assigns roles/Sites and approved services; creates Sites.
Observer	Reads authorised data; does not create Sites or execute control.
Operator / energy manager	Operational and strategy actions under explicit permissions.
Integrator	Configuration and integration within assigned scope.

Practical use

The member table includes name/email, role, Sites, services and actions. The editor shows current assignments. Diagnostic access to a customer Site requires its administrator’s approval, a limited duration, read-only scope and immediate revocation.

05 / FULL PRODUCT SCOPE

Services: requests, grants and suspension

Services; Settings / Users

Organisation entitlement and individual entitlement are separate. Administrators can grant a service without a prior request. The action takes effect without a second acceptance by the recipient.

Element / setting	Description and use
Organisation entitlement	Platform administrator selects an organisation and grants service/price zone.
Personal entitlement	Organisation administrator grants access to a selected person, including themselves.
Member request	Appears for a decision by the member’s administrator.
Organisation request	Organisation administrator requests access from the platform administrator.
Cancel / stop	Cancel a pending request or terminate a personal active grant.

Practical use

Approved for the organisation and Enabled for you identify different levels. Revoking an organisation service terminates dependent personal access; re-enabling does not automatically restore old personal grants. The catalogue includes day-ahead prices, charts, analytics, weather and forecasting; active modules follow the agreed scope.

06 / FULL PRODUCT SCOPE

Sites and energy balance

Sites / Overview

A Site combines location, infrastructure, assets and permissions. Monitoring shows generation, load, grid exchange and battery flows in a shared time context.

Element / setting	Description and use
Identification	Name, organisation, location/time zone and linked authoritative record.
Inventory	At least one infrastructure link; energy-asset inventory.
Current power	Generation, load, import/export, charge/discharge with units and sign.
Period energy	Integrated kWh/MWh quantities for the selected period.
Quality	Measurement time, freshness and source state.

Practical use

An administrator creates a Site. Members work with assigned Sites. Zero means measured zero; an unavailable source is displayed separately. Period comparisons use consistent units, time boundaries and aggregation rules.

07 / FULL PRODUCT SCOPE

Battery and BESS

Energy assets / Battery

The battery profile combines technical parameters, current state and economic settings. Control applies both Site limits and current BMS/PCS constraints.

Element / setting	Description and use
Capacity and power	Nominal/usable capacity; permitted charge/discharge power.
SOC / SOH	Charge level and BMS health estimate.
Operating range	Minimum, maximum and target SOC; emergency reserve.
Grid exchange	Separate permissions for grid charging and export discharge.
Measurements	Power, voltage, current, temperature, energy counters and alarms.

Practical use

The reference SunStorage PRO STE-261L-125P is an AC-coupled system rated at 125 kW and 261 kWh. These values describe that model; each Site is configured from its equipment documentation. The operating range does not raise factory limits.

08 / FULL PRODUCT SCOPE

Inverters and generation

Energy assets / Inverter

The inverter profile links generation measurements to the Site energy balance. Surplus use follows the mode and priorities for battery, loads, export and curtailment.

Element / setting	Description and use
Technical profile	Manufacturer, model, serial number, driver and communication link.
Telemetry	AC/PV power, voltages, currents, frequency, energy and status.
Surplus priority	Battery, flexible loads, export or curtailment.
Grid limit	Control against PCC measurement and permitted export maximum.

Practical use

At negative prices, the sale policy is considered alongside local demand, the battery and inverter-curtailment capabilities. Measurement and command support are specified for the exact model and protocol.

09 / FULL PRODUCT SCOPE

Charging and flexible loads

Energy assets / Charging station; Consumer and load

A flexible load describes consumption that can be scheduled within power, priority and deadline constraints. Chargers and industrial loads share Site-wide limits.

Element / setting	Description and use
Power	Minimum and maximum permitted power in kW.
Energy target	Required energy in kWh and completion deadline.
Priority	Load ordering when resources are limited.
Charger state	Connector, current power, session energy and total energy.
Integration option	Specific charger profile; OCPP as a separate project interface.

Practical use

Example: 30 kWh by 07:00, an 11 kW limit and a four-hour low-price window. Ideal minimum duration at that power is about 2.73 hours; the actual plan accounts for losses, charger limits and simultaneous loads. Price preference does not override the deadline.

10 / FULL PRODUCT SCOPE

Strategies: purpose and boundaries

Eight modes, one control framework

The catalogue includes intelligent hybrid, price arbitrage, self-consumption, zero export, peak shaving, schedule following, backup reserve and manual control. Mode shows the selected Site strategy. Activation follows a separate validation, approval and application process.

Unified configuration model

The Site strategy is a versioned document containing mode, parameters, inputs and constraints. The workflow separates preparation, validation, simulation, approval and activation. The audit trail links each change to user, time and revision. Physical control remains subordinate to local safeguards.

Shared settings

SOC limits are within 0-100%, minimum is below maximum and reserve stays inside them. The forecast horizon is 1-168 hours; the baseline profile uses 72 hours. Cycle projection covers the next 24 hours. Price-driven strategies specify two named forecast sources, their versions and weights. Their selection belongs to forecast-service configuration.

11 / FULL PRODUCT SCOPE

Energy and price strategies

Intelligent hybrid

Purpose: combine prices, PV production, load and battery charge. Inputs include current measurements, forecasts, tariffs, reserve and technical limits. Settings define the forecast horizon and target SOC before expected peaks. The intended output is a charge/discharge plan, not guaranteed savings.

Price arbitrage

Purpose: charge during economically favourable intervals and discharge at a positive net margin. The spread between two wholesale prices alone is insufficient: charge energy, conversion losses, fees, imbalance risk and selected battery costs matter. Parameters include price thresholds, minimum margin, reserve and limits. The margin rule applies to the selected price scenario.

Self-consumption

PV first serves local demand; surplus may charge the battery, which may support demand during a deficit within permitted limits. Consistent PV, PCC, load and battery measurements are required. Reserve and permitted import are Site-specific. The configuration is linked to the specific Site topology.

12 / FULL PRODUCT SCOPE

Grid and schedule strategies

Zero export

Purpose: keep export at the connection point below a configured limit using permitted charging, load control or generation curtailment. A reliable PCC meter with validated sign and freshness is required. A full battery, communication delay or unavailable PV control limits performance. This is not a claim of certified export protection.

Peak shaving

Purpose: limit peak demand against a contractual or technical threshold. Settings include threshold, reserve and pre-charge. Sufficient battery power and stored energy are required. Billing impact depends on the actual contract; not every tariff charges for peak demand.

Schedule following

The intended function follows an approved trading schedule and accounts for interval deviations. It requires schedule version, time zone, PCC measurements and imbalance rules. A 15-minute day normally has 96 intervals, or 92/100 at daylight-saving transitions. Schedule and balancing link to the trader’s contractual profile.

13 / FULL PRODUCT SCOPE

Reserve, manual control and safety

Backup reserve

The mode retains SOC for critical periods and loads. Reserve reduces energy available for arbitrage or self-consumption. Actual islanding, automatic transfer and UPS operation depend on the PCS, electrical design and protection equipment; a software mode cannot create these capabilities.

Manual control

A manual power request requires a separate permission, action reason and limited validity period. BMS limits and interlocks retain priority. The screen distinguishes requested, confirmed applied and actual measured power.

Final protective boundary

Edge control includes local limit checks, heartbeat/timeouts and driver-level clamping. Lost connectivity must not be presented as continuing confirmed control. Tests cover SOC bounds, charge/discharge limits, heartbeat loss, expired commands and safe fallback. Seeing a chart does not grant an observer control permission.

14 / FULL PRODUCT SCOPE

Strategy lifecycle

Draft, validation and simulation

A draft starts from a specific base version. Each save checks revision and protects concurrent edits. Checks cover parameters, battery availability, connection-point metering and required Site settings. Simulation uses a selected period up to 168 hours and retains input versions for reproducible comparison.

Activation → confirmation

Activation requires the relevant permission, writes enabled after commissioning, the correct revision and an idempotency key. Configuration is written to the linked Strategy Asset. An activating response means a request, not completed application. Status separates desired/applied revisions, control readiness, valid limits and Edge connectivity.

History and operational state

History shows versions, author, time, selected mode and activation result. Operators compare desired and applied revisions and see the reason for a rejection. Draft, Validated, Activating and Active identify separate steps. Returning to an earlier configuration follows the same permissions and checks.

15 / FULL PRODUCT SCOPE

Economics and loss protection

Two independent energy paths

Calculations distinguish direct PV sale from battery discharge. Direct PV sale is not charged for a battery cycle. Battery discharge includes source-energy cost, conversion loss and selected degradation/depreciation costs. Common components include trader/exchange fees, export fees, imbalance risk and minimum margin.

Cash cost or full cost

Cash-cost policy accounts for current expenses. Full-cost policy adds enabled depreciation and degradation. Minimum sale price = total cost + minimum margin. Net margin = sale price − total cost. A separate negative-price export block is enabled by default in validation. The result determines economic sale eligibility; local constraints are checked separately.

Illustrative example, not a tariff

At an 80 EUR/MWh charge price and 90% efficiency, calculated conversion cost is 8.89 EUR/MWh. With another 10 EUR/MWh in costs and a 5 EUR/MWh minimum margin, the threshold is 103.89 EUR/MWh. A sale at 120 gives a calculated margin of 21.11 EUR/MWh before any additional unapplied components. This is not a return forecast or customer quotation.

16 / FULL PRODUCT SCOPE

Battery cycles, provenance and tariffs

Equivalent full cycles

The projection function accepts usable capacity, grid charge, PV charge and discharge. Equivalent full cycles = (grid charge + PV charge + discharge)/(2 × usable capacity). Charging from each source divided by capacity is also reported separately. This is an energy-throughput metric, not a direct measurement of cell life loss.

Energy provenance

The energy ledger separates PV-to-battery, grid-to-battery, battery-to-load and battery-to-grid flows. Calculations use common intervals and consistent PV/PCC/load/battery measurements. Price, forecast, tariff and battery-configuration versions are linked to each economic scenario. This prevents the same charge being counted twice as a physical cycle.

Tariffs and contracts

Location: Settings / Market. Platform and organisation administrators manage authorised DSO and trader contract settings. Tariff, grid limits, balancing and trader schedule have separate versions. Wholesale price is stored separately from the final tariff. Reports distinguish energy cost, fees, forecast outcome and a financially closed period.

17 / FULL PRODUCT SCOPE

Prices: delivery, precision and archive

When prices apply

Day-ahead means a price for a specific future delivery interval, not the price at download time. The display must identify delivery day and Europe/Sofia time zone. ENTSO-E A44 is the market-price source; tomorrow’s availability depends on the provider’s publication. Missing intervals are not filled with zero or yesterday’s price.

Granularity and refresh

BG retains native 15-minute intervals; hourly values are a compatible aggregation, not a replacement for originals. The documented worker checks tomorrow hourly until complete and recovers the current day at startup/day rollover. An hourly request does not imply hourly market resolution. Last API check, last successful complete import and last priced interval are separate.

Country management and access

Only Bulgaria is collected by default. Another country/bidding zone requires explicit platform-administrator enablement. The archive and revisions are retained in a separate TimescaleDB without automatic deletion. The full analytical archive and provider diagnostics are platform-admin functions; the customer view follows service, zone and personal entitlements.

Charts, history and forecasts

Two distinct visualisation sources

The embedded BG Grafana view uses authorised market data. Site charts obtain configured OpenRemote measurement history through the GrideX API. Entitlement to one service does not open arbitrary databases or foreign Sites. Market periods include today, tomorrow, 48 hours, 7/30 days and a custom range up to 31 days.

Measurement history

Measurement, publication and heartbeat use separate periods. History stores accepted timestamped events for selected metrics. Profile application is tracked through device acknowledgement. Missing values and old measurements retain their actual status and time. An empty period is labelled as no data.

Weather, forecasting and analytics

These modules form part of the full project specification for services. Weather inputs include temperature, solar radiation, cloud cover and wind. Forecasting combines these inputs with generation, load and price history. Analytics compares predicted and actual values, baseline and selected schedule. Every result carries a period, version and unit.

19 / FULL PRODUCT SCOPE

Infrastructure and provisioning

From selection to confirmed asset

An administrator creates a Site and selects approved infrastructure/device with a specific model and role. The backend checks organisation ownership and creates/links the corresponding OpenRemote record. Pending or failed provisioning is not presented as a working asset. Repeating an operation must not create a second independent Site.

Hardware profile

The reference ROCK/ESP configuration includes one ROCK Pi E and ESP nodes with one CAN or isolated RS485 device port each. Drivers are defined by type, model and protocol. Model compatibility covers registers, scaling, sign convention and permitted commands. Catalogue extension follows validation of the exact hardware and firmware version.

Settings and secrets

Configuration sections cover battery, tariff, forecast, grid, charger, notifications, trader schedule and balancing. Revisions protect against conflicting edits. Passwords, keys and connection details must not be returned as public attributes. Changing a desired setting and actual Edge application are separate events.

20 / FULL PRODUCT SCOPE

Connectivity, alarms and notifications

Connection state

Heartbeat is separate from PCS/BMS measurements. ROCK receipt time, successful ESP contact and measurement time are not interchangeable. Documented defaults are 30 seconds for stale and 90 for offline; these are settings, not an SLA. A never-seen device is unknown, not operational.

One-shot notification

Users opt in through their profile for future incidents at accessible Sites. A verified email and renewed access check are required before sending. At most one message is sent per offline episode; recovery permits a new incident. Late opt-in does not send retrospective mail. An offline ROCK does not generate duplicate messages for each dependent ESP.

Delivery and different message types

Invitations, service decisions, connectivity alerts and customer enquiries are separate processes. Provider outcomes and uncertain states are tracked to limit repeated messages. The enquiry form is protected against automated abuse. Alarm is located under Mode.

21 / FULL PRODUCT SCOPE

Architecture, operations and support

Component responsibilities

The API handles identity, navigation, permissions, Sites, settings, strategies and protected entry. Keycloak manages identities/passwords; OpenRemote manages inventory and measurements. Separate processes collect market prices, heartbeats and history, and send warnings. MQTT serves authorised devices; proxies enforce public boundaries; Grafana visualises authorised data.

Operational checks

Operational checks cover sign-in/out, organisation isolation, Site access, services, current measurements, price intervals, notifications and restart recovery. Results are tracked per component and end-to-end service. Provider and customer define response and recovery times in the agreed support plan.

Deployment and support

The architecture separates portal, identity, API, asset inventory, time-series data, processing workers and local control. Hosting is selected according to data volume, Site count, availability and backup policy. A cloud virtual machine, including Google Cloud, is a deployment option; resources and recovery are sized for the project.

22 / FULL PRODUCT SCOPE

Forecast service and scenarios

Services / Forecasting - project module

The forecast specification connects three time series: generation, consumption and market price. The baseline horizon is 72 hours in 15-minute intervals. Forecast output feeds optimisation rather than acting as a direct command.

Element / setting	Description and use
Generation	PV power from history, irradiance, temperature and cloud cover.
Demand	Load with calendar factors, history and production plan.
Price	Wholesale price and forecast source, separate from final tariff.
Method	Specified LightGBM model; separate target for each series.
Traceability	Model version, period, input versions and confidence estimate.

Practical use

Inputs include lags of 15 minutes, 1 hour, 24 hours and 7 days, 1/24-hour averages and calendar features. Model validation follows chronological order to avoid using future data when evaluating an earlier period. Forecast and realised returns are shown separately.

23 / FULL PRODUCT SCOPE

Control and forecast parameters

Mode / Logic

Parameters define strategy behaviour and acceptable input age. The ranges below come from the functional contract; effective settings are further constrained by equipment and Site policy.

Element / setting	Description and use
Control interval	1-900 seconds; separate from measurement and publication.
Re-optimisation	1-1440 minutes; recalculation interval.
Forecast horizon	1-168 hours; input maximum age 1-1440 minutes.
Confidence	Threshold 0-100%; condition for using a forecast.
Pre-charge	1-72-hour horizon for peak shaving.
Ramp and fallback	Ramp in kW/min; safe idle, self-consumption or schedule following.

Practical use

Example rule: when tomorrow’s expected solar production is low, preserve 80% SOC and permit grid charging only below the economic threshold. 80% is an example setting, not a universal recommended reserve. User configuration does not alter BMS protective limits.

24 / FULL PRODUCT SCOPE

Tariff model and period calculation

Settings / Market / Tariff and settlement

A tariff profile combines energy price with applicable contractual components. Purchase and sale, quantity and unit price, forecast calculations and final accounting periods remain separate.

Element / setting	Description and use
Version and validity	Name, contract reference, validity dates and currency.
Purchase / sale	Separate prices or contractual links to a market index.
Fees	Trader, exchange, network and export charges per contract.
Imbalance	Selected risk/cost and applicable contractual method.
Period result	Export revenue minus import and included costs.

Practical use

For variable pricing, cost is summed by interval: MWh energy multiplied by the matching EUR/MWh price. For example, 10 kW average power over 15 minutes is 2.5 kWh, or 0.0025 MWh; at 100 EUR/MWh the energy component is EUR 0.25. Taxes and other contractual components are added only when applicable and configured. The calculation does not replace the supplier’s invoice.

25 / FULL PRODUCT SCOPE

Balancing and production context

Settings / Market / Balancing; Services / Analytics

The project balancing module compares the contracted schedule with the Site’s measured position. Industrial integrations add expected production-plan demand to the load forecast.

Element / setting	Description and use
Schedule	Identifier, version, delivery date, time zone and interval positions.
Deviation	Measured energy versus schedule under a shared sign convention.
Tolerance	Permitted power deviation and schedule-adherence weight.
ERP / MES / WMS	Project connections for orders, shifts, loads and energy models.
VPP / DSO / communities	Separate project interfaces, contracts and participation rules.

Practical use

Comparable reporting retains data provenance and schedule version. Forecast imbalance prices and final prices are distinct datasets. Balancing-market participation, aggregation or an energy community depends on the relevant operator and contract, not only on available measurements.

26 / FULL PRODUCT SCOPE

Product extensions and configurations

Full project catalogue

Extensions complement the core energy model through defined integration and contractual scope. They are product options that preserve isolation, access and safety rules.

Element / setting	Description and use
Additional drivers	Additional models with specified protocol, registers, units and commands.
Cloud connectors	External-provider connections with scoped permissions and traceable sources.
OCPP / portfolios	Charging integrations and aggregation of agreed Sites.
Partner model	Project variants for partners, white-label and licensing scope.
Languages	BG/EN; extension through translation catalogues and locale formatting.
Plan and subscription	Contracted services/resource package; pricing and billing by quotation.

Practical use

Existing illustrative commercial values are not a customer quotation. Country-based suggestions, additional languages, payment channels and partner branding are defined in the relevant project specification. A user’s language choice is not changed merely because they use a VPN.

27 / FULL PRODUCT SCOPE

Updates, continuity and data

Infrastructure; operations plan

Configuration lifecycle is separate from firmware lifecycle. Every change has a version, target device, execution permission and recorded outcome.

Element / setting	Description and use
Configuration	Draft, approval, retrieval, application and readback.
Edge update	Controlled delivery; ESP devices are serviced through ROCK.
Local journal	Disk-bounded telemetry buffer.
Recovery	Project process for transfer, acknowledgement and duplicate handling.
Backups	Databases, configuration and protected keys; retention and recovery test.

Practical use

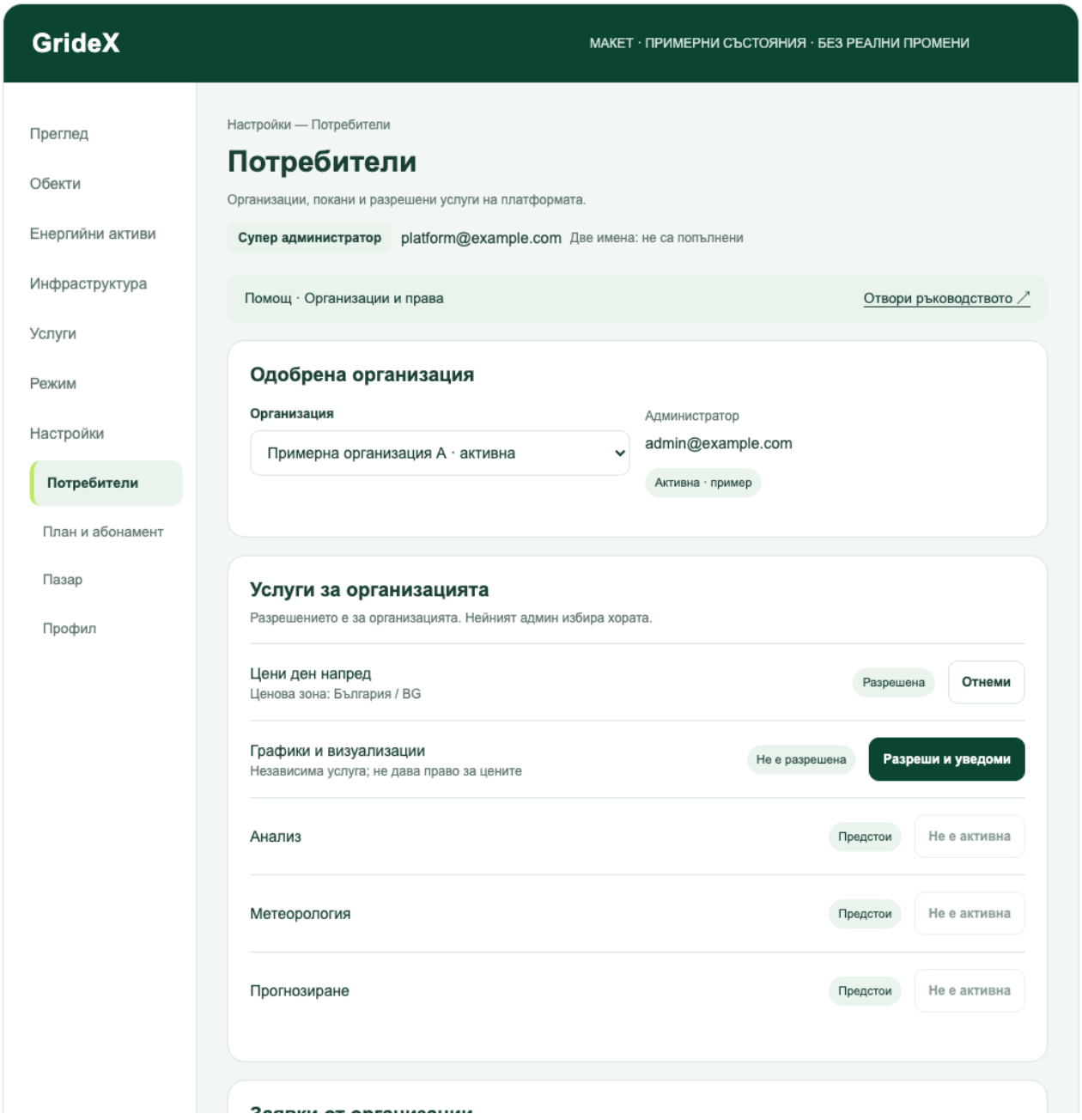
During a connection loss, data state remains visible. Unreceived measurements are not replaced with fabricated values. Local retention duration, acceptable data loss and recovery time are sized for the Site. Measurement backups and the permanent market-price archive have separate policies.

Super administrator

Approved reference mockup from the documentation, shown in Bulgarian. All identities and Sites are examples.

Part 1 of 2 · Settings → Users

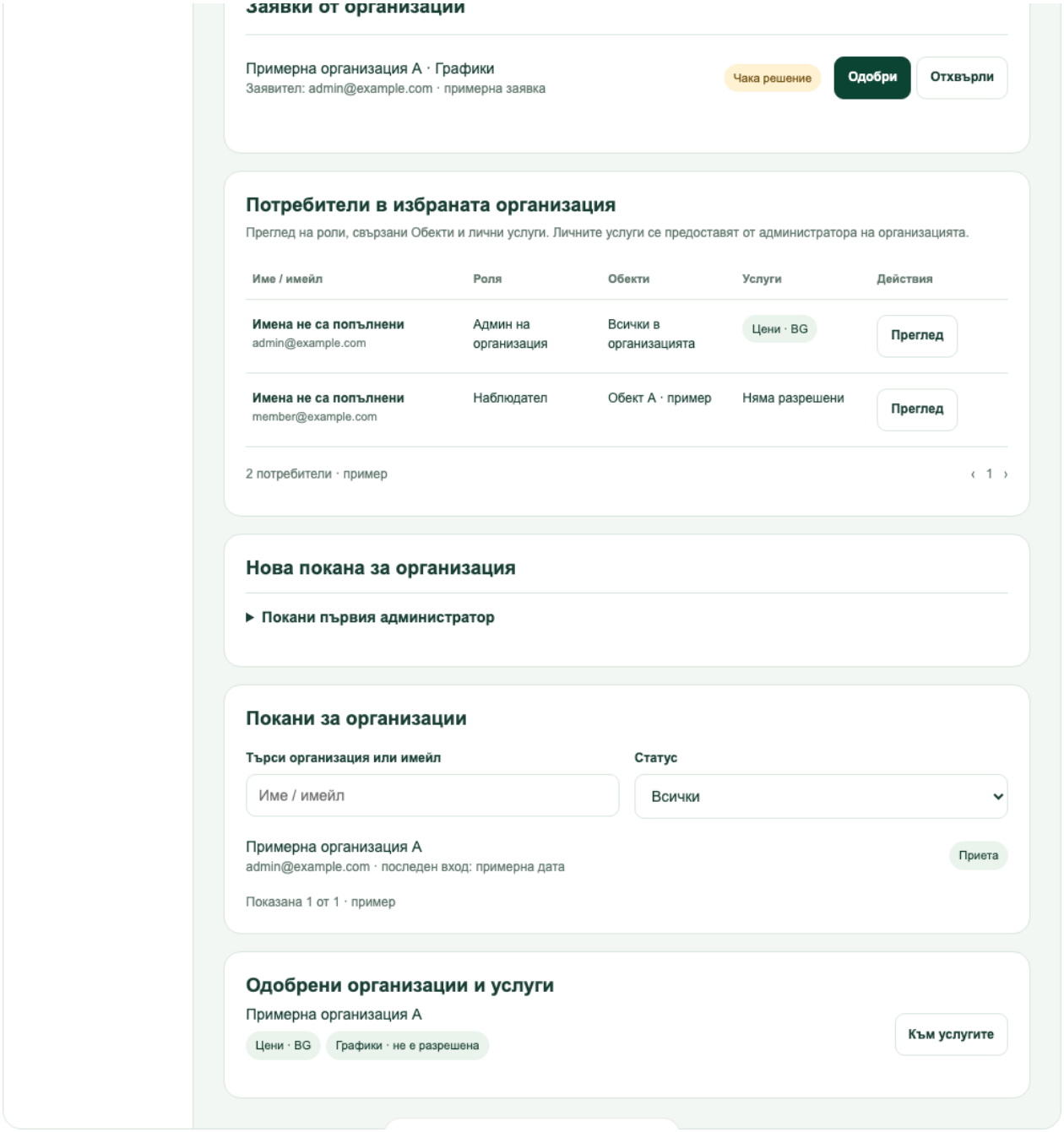
Супер администратор · организации и организационни услуги



Super administrator

Approved reference mockup from the documentation, shown in Bulgarian. All identities and Sites are examples.

Part 2 of 2 · Settings → Users

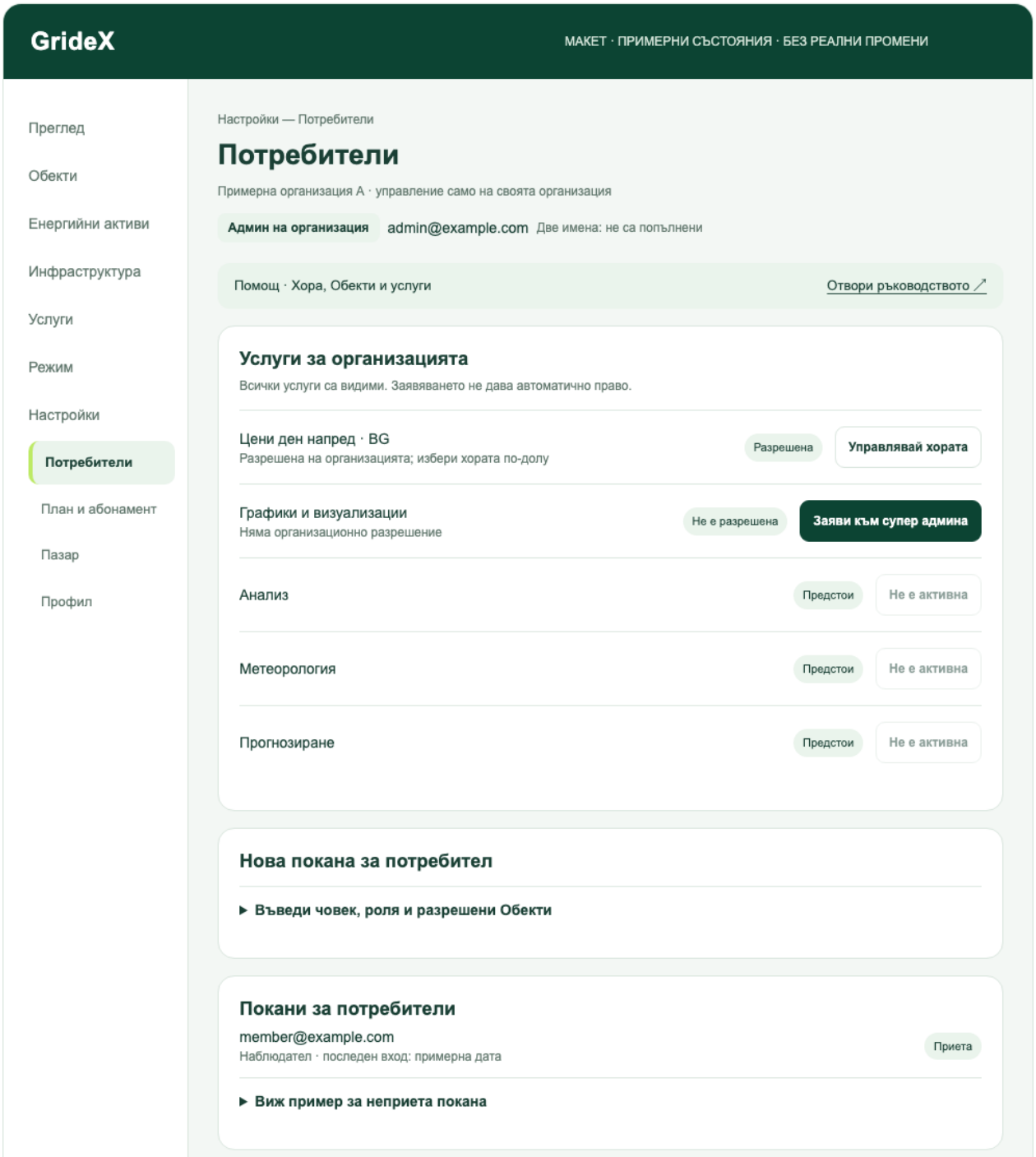


Organisation administrator

Approved reference mockup from the documentation, shown in Bulgarian. All identities and Sites are examples.

Part 1 of 2 · Settings → Users

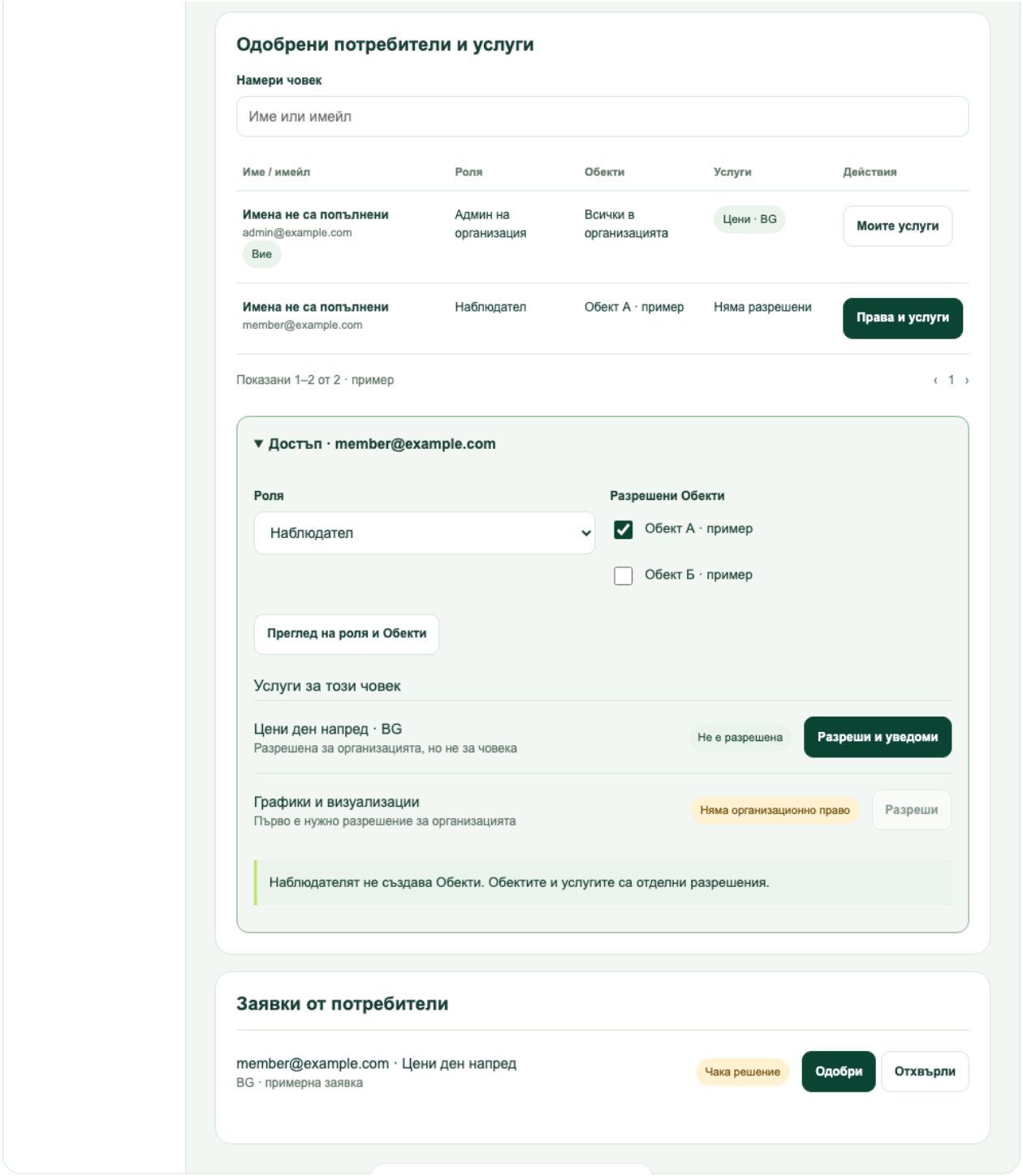
Администратор на организация · конкретни хора, Обекти и услуги



Organisation administrator

Approved reference mockup from the documentation, shown in Bulgarian. All identities and Sites are examples.

Part 2 of 2 · Settings → Users

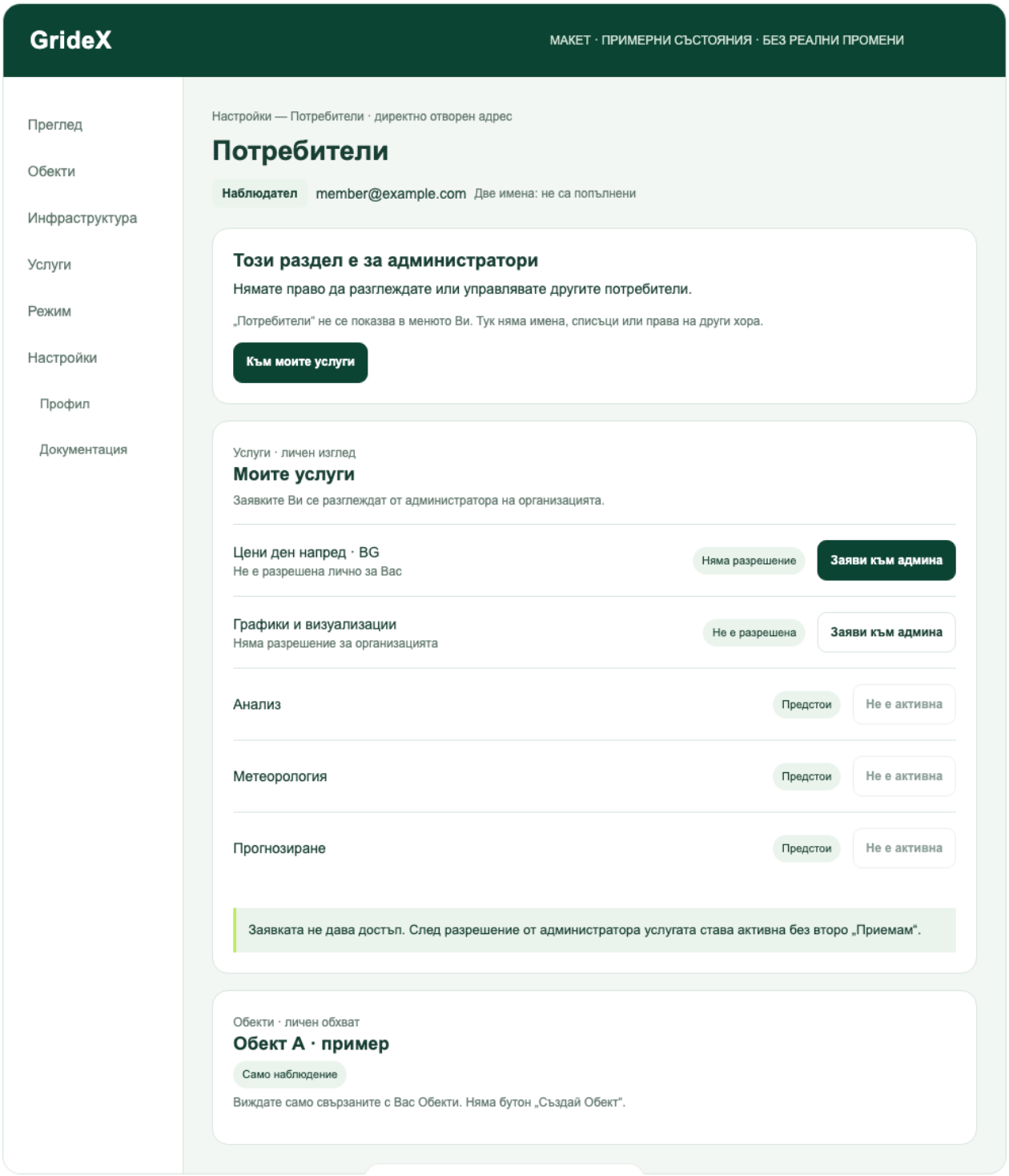


Member personal access

Approved reference mockup from the documentation, shown in Bulgarian. All identities and Sites are examples.

Part 1 of 1 · Settings → Users

Потребител · личен достъп, без административен регистър



Terminology, documentation and contact

Term	Meaning
SOC	State of charge
SOH	State of health
PCC	Point of common coupling
BMS / PCS	Battery management / power conversion system
kW / kWh	Power / energy
EFC	Equivalent full cycles
ERP / MES / WMS	Business / manufacturing / warehouse systems

Help and support

gridex.tech
doc.gridex.tech
support@gridex.tech

The specification is based on the GrideX product model: navigation and permissions, strategy/settings contract, market services, Edge integration and forecast module. Examples explain the method; commercial terms and warranties are contractual.

Suntech STE-261L-125P: the 125 kW / 261 kWh ratings follow manufacturer product information. GrideX is an open-source project (MIT). Trademarks belong to their respective owners.